

NTOU English FinalTerm Show-me-How TEST

[OPENING]

Hey, Guys!

Have you ever thought about how you usually count on your hands?

You probably do it something like this, for example:

[--ACTUAL Show ST--]

1, 2, 3, and to the 10.

(實際用手來演示怎麼比1、2、3、10)

[--ACTUAL Show ED--]

Or, some of you smart folks might think of using one hand for the tens digit and the other for the ones digit.

This way, we can keep counting all the way up to 99 (ninety-nine). For instance:

[--ACTUAL Show ST--]

21 (twenty-one), 67 (sixty-seven).

(實際用雙手來演示怎麼比這4個數字)

[--ACTUAL Show ED--]

But obviously, none of us here is a thousand-handed Guanyin or an octopus, so the biggest number you can show this way is 99 (ninety-nine).

Today, I want to show you a brand-new way to count. Using just your ten flexible fingers, you can count past 100 (one hundred), or even over 1000 (one thousand)!

[Introduction]

The system we normally use, where we carry over after 9 (nine), is called the decimal system.

But today, to max out the potential of every single finger, we are going to use the binary system to count!

As the name suggests, in the binary system, the number 2 (two) doesn't exist.

You carry over whenever you pass 1 (one).

For example, "1 (one) 0 (zero)" in binary doesn't mean 10 (ten); it actually represents the number 2 (two).

And we can use our fingers—putting them up or keeping them down—to represent 1 and 0. You smart cookies will definitely notice that this rule is closely related to the powers of 2 .

[--ACTUAL Show ST--]

Imagine your rightmost finger is 2 to the power of 0

Every time you move one finger to the left, the power goes up by 1.

Following this logic, your right index finger would be 2 to the power of 1, and your left pinky would be 2 to the power of 5 .

(實際演示哪支手指代表 2^0 、哪支代表 2^1 、哪支代表 2^5)

[--ACTUAL Show ED--]

The fingers further to the right represent smaller numbers.

The rightmost bit is called the "Least Significant Bit," or LSB.

Conversely, the fingers further to the left represent larger numbers.

The leftmost bit is called the "Most Significant Bit," or MSB.

[--ACTUAL Show ST--]

To make this easier to show,

I'm going to put sticky notes on the LSB and MSB positions so it's easier to understand.

(在手上貼上便利貼來註記)

[--ACTUAL Show ED--]

Since we have 10 fingers in total, a quick calculation shows that with this method, our hands can represent up to 2 to the power of 10

which gives us 1024 (one thousand and twenty-four) different numbers!

[--ACTUAL Show ST--]

Making a fist with both hands means all 10 bits are 0, which represents the decimal number 0 .

Holding all your fingers up means all 10 bits are 1 , which represents the decimal number 1023 (one thousand and twenty-three).

(實際用雙手來演示)

[--ACTUAL Show ED--]

Besides showing the maximum and minimum values,

I' ll now demonstrate a few specific numbers. For example:

[--ACTUAL Show ST--]

819 (eight hundred and nineteen) in decimal becomes "11001 10011" in a 10-bit binary, which looks like this.

1009 (one thousand and nine) in decimal becomes "11111 10001" in a 10-bit binary, which looks like this.

(對每個數字都實際用雙手來演示)

[--ACTUAL Show ED--]

You guys can also try doing this with your own hands and see what those numbers translate to in decimal!

[END]

I hope this fun way of counting brings a smile to your face.

Thank you everyone, that concludes my show!